

Product Service API Documentation

REST API Design & Professional API Development — Epic 03

1. Overview

Product Service is a RESTful microservice responsible for managing product information.

Service Name: Product Service

Base URL: http://localhost:8082

API Version: v1

Data Format: JSON

2. Product Resource

Field	Type	Description
id	Long	Unique product ID
name	String	Product name
description	String	Product description
price	Number	Product price
category	String	Product category
stock	Integer	Available stock
status	String	Product availability status
createdAt	LocalDateTime	Product creation time
updatedAt	LocalDateTime	Product last update time

3. API Endpoints

3.1 Get All Products

Request: GET /api/v1/products

Success Status: 200 OK

Example: [{"id":1,"name":"Laptop","description":"Dell Laptop","price":55000,"category":"Electronics","stock":10,"status":"AVAILABLE","createdAt":null,"updatedAt":null}]

3.2 Get Product By ID

Request: GET /api/v1/products/{id}

Success Status: 200 OK

Example: {"id":1,"name":"Laptop","description":"Dell Laptop","price":55000,"category":"Electronics","stock":10,"status":"AVAILABLE","createdAt":null,"updatedAt":null}

3.3 Create Product

Request: POST /api/v1/products

Success Status: 201 Created

Example: {"name":"Laptop","description":"Dell Laptop","price":50000,"category":"Electronics","stock":10,"status":"AVAILABLE"}

3.4 Complete Product Update

Request: PUT /api/v1/products/{id}

Success Status: 200 OK

Example: {"name":"Samsung Galaxy","description":"Updated Samsung Mobile","price":30000,"category":"Electronics","stock":15,"status":"AVAILABLE"}

3.5 Partial Product Update

Request: PATCH /api/v1/products/{id}

Success Status: 200 OK

Example: {"price":55000}

3.6 Delete Product

Request: DELETE /api/v1/products/{id}

Success Status: 204 No Content

Example: No response body

4. HTTP Status Codes

Status Code	Meaning
200 OK	Request completed successfully
201 Created	Product successfully created
204 No Content	Product successfully deleted
400 Bad Request	Invalid request or validation failure
404 Not Found	Product does not exist
409 Conflict	Resource conflict
500 Internal Server Error	Unexpected server error

5. Validation

The Product Service validates product information before processing requests. Invalid input returns 400 Bad Request.

Example: Product price must be greater than or equal to 0

6. Error Handling

Product Not Found

Request: GET /api/v1/products/9999

Status: 404 Not Found

Response: Product not found with id: 9999

7. Postman Testing

The following scenarios were tested using Postman:

- Create Product
- Get All Products

- Get Product By ID
- Complete Product Update
- Partial Product Update
- Delete Product
- Invalid Product ID
- Non-existing Product
- Invalid/Missing Product Fields

The tested scenarios return the appropriate HTTP status codes.

8. API Summary

Operation	HTTP Method	Endpoint	Success
Get all products	GET	/api/v1/products	200
Get product	GET	/api/v1/products/{id}	200
Create product	POST	/api/v1/products	201
Complete update	PUT	/api/v1/products/{id}	200
Partial update	PATCH	/api/v1/products/{id}	200
Delete product	DELETE	/api/v1/products/{id}	204

9. Service Port

Product Service → **8082**

Base URL: `http://localhost:8082`

10. API Version

All Product Service APIs use version 1: `/api/v1/products`. This allows future API versions to be introduced without breaking existing clients.